

BASE ESPERANZA, Antarctica

WMO: 889630

Lat: 63.3989S

Lon: 56.9978W

Elev: 79

StdP: 14.65

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-12.9	-8.9	-22.7	1.6	-11.0	-18.4	2.0	-7.6	73.9	2.9	68.5	0.8	25.8	230	1.158

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.4	47.4	39.7	44.9	38.2	42.9	37.1	40.5	45.5	39.1	43.6	37.8	41.8	21.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
36.6	31.9	40.3	35.4	30.4	38.9	34.5	29.3	37.9	15.6	45.7	14.9	43.9	14.4	42.0	51.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 59.0	50.3	45.4	DB	-16.2	53.5	4.4	3.1	-19.3	55.8	-21.9	57.6	-24.3	59.3	-27.5	61.5	(4)
(5)			WB	-16.6	44.8	4.1	2.5	-19.6	46.6	-22.0	48.1	-24.3	49.5	-27.3	51.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.8	34.7	33.9	28.8	21.2	18.7	13.5	11.6	14.6	19.6	25.1	30.2	33.8
	DBStd	12.38	3.46	5.25	7.99	10.51	10.82	12.15	12.72	11.75	11.92	9.70	5.72	3.68
	HDD50	9577	474	450	657	864	970	1094	1189	1098	911	772	594	504
	HDD65	15052	939	870	1122	1314	1435	1544	1654	1563	1361	1237	1044	969
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.3	12.7	14.9	16.3	17.1	15.9	17.5	17.5	17.7	18.5	18.0	15.9	14.1
Precipitation	PrecAvg	20.3	1.9	1.7	2.2	1.7	1.7	1.5	1.4	1.7	1.8	1.7	1.8	1.6
	PrecMax	47.3	5.5	4.8	4.8	4.3	4.3	3.7	3.5	6.0	8.1	7.2	5.7	4.3
	PrecMin	7.1	0.1	0.0	0.1	0.2	0.0	0.1	0.0	0.0	0.0	0.3	0.2	0.2
	PrecStd	9.0	1.3	1.1	1.2	1.0	1.1	0.9	0.9	1.5	1.6	1.4	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.5	50.3	49.8	44.6	49.3	42.1	41.9	40.0	44.3	45.5	46.8	47.0
		MCWB	42.1	41.5	41.5	37.5	40.1	34.8	35.4	34.2	37.0	37.5	39.4	39.1
	2%	DB	45.1	46.3	44.6	40.4	41.1	37.4	36.0	35.7	40.0	41.8	42.9	43.7
		MCWB	38.8	38.9	38.0	35.5	35.6	32.5	32.4	32.2	34.5	36.0	36.8	37.5
	5%	DB	42.6	44.0	41.3	37.5	36.7	34.2	33.3	33.3	37.3	39.2	40.2	41.3
		MCWB	37.4	37.9	36.4	34.1	33.3	31.7	31.6	31.2	33.3	34.4	35.3	36.0
	10%	DB	40.6	41.7	38.9	35.4	34.2	31.6	31.1	31.5	34.8	36.8	37.9	39.5
		MCWB	36.2	36.6	35.2	33.0	32.1	30.1	29.8	29.8	32.3	33.3	34.0	34.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.8	42.6	42.3	39.1	41.1	35.6	35.9	35.1	38.1	39.2	40.4	40.4
		MCDB	47.7	49.5	48.8	42.7	48.3	39.6	40.2	37.9	43.0	43.2	45.5	44.7
	2%	WB	39.6	39.7	38.8	36.4	36.1	33.4	33.1	33.0	35.1	36.7	37.7	38.3
		MCDB	43.8	44.8	43.1	39.3	40.0	36.1	34.9	34.8	38.4	40.8	41.9	42.5
	5%	WB	38.0	38.5	37.1	34.6	33.9	31.9	31.6	31.5	33.8	35.0	35.9	36.7
		MCDB	41.6	42.9	40.1	36.9	36.3	34.1	33.3	33.1	36.5	38.2	39.2	40.3
	10%	WB	36.6	37.2	35.7	33.0	32.1	30.2	29.8	29.9	32.5	33.6	34.4	35.5
		MCDB	39.9	41.1	38.2	35.1	34.1	31.7	31.4	31.5	34.7	36.2	37.2	38.6
Mean Daily Temperature Range	5% DB	MDBR	7.4	8.3	9.3	10.0	11.9	11.6	12.5	12.6	12.4	10.8	8.6	7.5
		MCDBR	11.3	13.2	12.0	13.1	15.9	14.3	15.7	15.1	13.9	13.0	12.2	10.9
		MCWBR	7.1	8.2	7.8	9.8	12.2	11.4	13.5	13.3	10.3	8.9	8.2	7.0
	5% WB	MCDBR	10.6	12.1	10.8	12.5	13.9	13.6	15.2	14.0	12.5	11.9	11.0	10.2
		MCWBR	7.2	8.1	7.8	10.0	11.1	11.8	13.6	12.6	10.0	9.0	8.2	7.3
Clear-Sky Solar Irradiance	taub	0.293	0.289	0.275	0.247	0.180	0.153	0.168	0.236	0.264	0.285	0.285	0.292	
	taud	2.419	2.434	2.439	2.344	2.032	1.923	1.989	2.221	2.301	2.320	2.367	2.375	
	Ebn at Noon	310	295	271	222	145	82	143	217	267	292	311	315	
	Edh at Noon	33	29	23	16	8	4	8	17	25	32	35	36	
All-Sky Solar Radiation	RadAvg	1816	1265	693	291	82	25	55	229	641	1221	1795	2058	
	RadStd	118	67	53	21	6	1	3	16	43	59	92	135	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+3.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page